

AL WAJH, Saudi Arabia

WMO: 404000

Lat: 26.199N

Lon: 36.476E

Elev: 20

StdP: 101.08

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	13.9	-1.0	3.5	19.1	1.1	4.1	18.9	12.2	20.0	10.8	20.7	2.7	20	0.442

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	36.1	25.2	35.0	26.7	34.1	27.2	30.2	33.4	29.7	33.1	29.2	32.7	5.4	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.2	26.0	32.6	28.9	25.6	32.5	28.1	24.4	32.0	101.8	34.0	98.8	32.7	96.2	32.8	33.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	10.2	9.3	10.4	42.3	1.7	2.3	9.1	44.0	8.1	45.4	7.2	46.7	5.9	48.4		
			5.1	31.3	1.8	0.9	3.9	31.9	2.8	32.5	1.8	33.0	0.6	33.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	19.6	20.4	22.2	24.7	27.5	28.7	30.2	30.5	29.2	27.6	24.7	21.4
	DBStd	4.29	2.19	2.29	2.55	2.51	2.38	2.03	1.30	1.29	1.50	1.80	2.19	2.18
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	18	11	5	1	0	0	0	0	0	0	0	0	2
	CDD10.0	5688	297	291	377	441	542	562	626	636	577	546	440	353
	CDD18.3	2664	49	63	120	191	284	312	367	377	327	288	190	96
	CDH23.3	31443	244	341	810	1727	3368	4141	5309	5520	4371	3362	1662	588
Wind	WSAvg	4.2	3.8	4.3	4.5	4.4	4.4	4.6	4.1	4.3	4.4	3.9	3.6	3.6
	PrecAvg	34	15	4	1	1	0	0	0	0	0	2	5	6
	PrecMax	78	59	52	10	4	4	1	0	0	3	17	37	55
	PrecMin	9	3	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	22	15	10	2	1	1	0	0	0	1	4	10	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	30.2	33.0	35.0	37.2	38.7	37.1	36.9	36.5	35.8	34.8	31.9
		MCWB	19.1	18.9	20.8	21.7	22.6	23.4	25.6	27.2	25.5	22.9	19.9	18.6
	2%	DB	27.0	28.0	30.0	32.2	34.8	35.1	35.2	35.2	34.2	33.2	31.9	29.0
		MCWB	19.4	20.0	20.7	22.2	23.7	25.4	27.2	28.4	27.5	26.3	22.4	19.8
	5%	DB	25.9	26.5	28.2	30.9	32.9	33.9	34.2	34.4	33.2	32.2	30.6	27.7
		MCWB	19.0	19.7	21.0	22.1	23.7	26.1	27.6	28.5	27.7	26.4	22.5	19.8
	10%	DB	24.8	25.2	27.0	29.2	31.8	32.8	33.8	33.9	32.9	31.6	29.4	26.2
		MCWB	18.1	18.8	20.7	22.1	24.1	26.1	27.7	28.4	27.7	26.1	22.1	18.8
	0.4%	WB	23.3	23.7	24.6	26.2	28.2	29.6	31.0	31.3	30.6	29.6	26.2	24.0
		MCDB	26.0	26.5	28.1	30.2	32.1	33.4	33.8	33.9	33.2	32.5	30.3	27.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.6	22.3	23.5	25.2	27.2	28.7	30.1	30.4	29.7	28.5	25.2	22.4
		MCDB	25.2	25.7	27.3	29.2	31.2	32.5	33.3	33.5	32.7	31.6	29.4	26.6
	5%	WB	20.0	21.1	22.5	24.4	26.5	27.9	29.4	29.8	29.1	27.7	24.1	21.0
		MCDB	24.4	25.2	26.6	28.4	30.5	31.6	32.9	33.3	32.3	31.1	28.8	25.9
	10%	WB	18.6	19.8	21.5	23.6	25.8	27.3	28.8	29.3	28.4	26.9	23.1	19.8
		MCDB	23.8	24.6	26.0	27.7	30.0	31.2	32.4	32.8	31.7	30.7	28.1	25.3
Mean Daily Temperature Range	5% DB	MDBR	9.4	9.4	9.3	8.7	8.1	7.8	7.5	7.0	7.6	8.3	9.6	9.7
		MCDBR	10.6	10.5	10.6	10.3	9.6	9.2	8.2	7.4	8.0	8.6	10.6	11.1
		MCWBR	8.0	8.8	9.0	9.0	8.7	7.8	6.4	5.8	6.9	8.4	8.6	8.2
	5% WB	MCDBR	9.3	9.1	9.2	8.5	8.1	7.8	7.5	6.8	7.4	8.0	9.6	9.6
		MCWBR	8.0	8.6	8.6	8.1	8.0	7.0	6.4	6.0	6.8	8.4	8.6	7.8
Clear-Sky Solar Irradiance	taub	0.370	0.415	0.458	0.520	0.513	0.461	0.485	0.513	0.472	0.466	0.406	0.382	
	taud	2.342	2.166	2.032	1.889	1.924	2.090	2.064	1.996	2.075	2.078	2.242	2.294	
	Ebn at Noon	874	856	838	794	797	834	814	791	814	793	825	841	
	Edh at Noon	110	141	170	201	194	164	168	179	161	152	121	111	
All-Sky Solar Radiation	RadAvg	4.29	5.28	6.55	7.38	7.91	8.30	8.02	7.52	6.79	5.63	4.62	4.03	
	RadStd	0.24	0.19	0.19	0.21	0.26	0.09	0.13	0.13	0.16	0.13	0.14	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	+0.61	N/A	N/A	+0.68	+0.60	+0.52	N/A	N/A	+231	+227	
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page